

Written Assignment - Lexical Analysis

1. Construct a DFA M , which can accept the a, b, c strings which begin with a or b , or begin with c and contain at most one a . Please write a C++ function to implement the DFA.

3. Please construct a **DFA with minimum states** for the following regular expression.

(1) $aa^*(bab^*a)^*(a|b)b^*$

(2) $a(bab^*a)^*(a|b)b^*$

(3) $(ab^*a)^*(a|b)b^*$

Complete the homework and submit it before the next next class to the teaching assistant (Room 442/444, Building of Sch. Computer Sci&Eng).

1. Everyone should give their completed workbook to the teaching assistant before the designated time. There is a box at the door of the room specifically for collecting homework.

2. After the first submission of each assignment, it will be graded based on the quality of completion. Allow the next submission to modify errors from the previous assignment. The final score for each assignment will be given based on the final quality and number of revisions made.